# Chapter 1 - Introduction to Probability Part A - Counting

## Selection of Distinguishable and Indistinguishable Objects

The number of ways to select subsets of size $r$ from a set of $n$ distinguishable or indistinguishable objects is given in the following table, summarizing the four different scenarios:

| | With replacement | Without replacement |
|---|---|---|
| Ordered | $n^r$ | ${}_nP_r = \dfrac{n!}{(n-r)!}$ |
| Unordered | not covered in this course | ${}_nC_r = \dbinom{n}{r} = \dfrac{n!}{r!(n-r)!}$ |

## Factorials

Number of arrangements of $n$ **distinguishable** objects

$$
\begin{aligned}
&= (\text{number of possible choice of the 1st object}) \\
&\qquad \times (\text{number of possible choice of the 2nd object}) \\
&\qquad\qquad \times (\text{number of possible choice of the 3rd object}) \\
&\qquad\qquad\qquad \times \cdots \\
&\qquad\qquad\qquad\qquad \times (\text{number of possible choice of the } n\text{-th object}) \\
&= n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1 \\
&= n!.
\end{aligned}
$$

Therefore, the number of permutations of a set of $n$ **distinguishable** objects, taken all together, is $n!$. The notation $n!$ is called the factorial of $n$. By convention, $0!$ is defined as 1.

**Example 1:**

(a) How many ways can 4 people sit in a row?

(b) How many ways can 4 people sit around a round table?

(c) There are 2 boys and 3 girls in a group. They go to the cinema and take a 5-seat row. How many different ways of seating are there, if Keith and Eric are required to sit together?

**Solution:**

(a) $4! = 24$.

(b) $4!/4 = (4-1)! = 6$.

(c) $4!2! = 24 \times 2 = 48$.

# Permutations - Ordered Selection without Replacement

**Definition:** An arrangement of $r$ objects taken from a set of $n$ **distinguishable** objects is called a permutation of $n$ objects taken $r$ at a time. The total number of such **ordered** subsets are symbolized ${}_nP_r$ $(r \leq n)$ and can be calculated by

$$ {}_nP_r = n(n-1)\cdots(n-r+1) = \frac{n!}{(n-r)!}. $$

**Equivalent Problem:** allocate $r$ distinguishable objects into $n$ distinguishable cells, with no cells containing $> 1$ objects.

**Example 2:** In how many ways can 8 different books be arranged on a shelf?

**Solution:** ${}_8P_8 = 8! = 8 \times 7 \times \cdots \times 1 = 40320$.

**Example 3:**

A beauty contest has 15 candidates. How many ways can awards be given to the winner, 1st and 2nd runners-up?

**Solution:**

Problem: $n$ distinguishable objects, choose $r$ from them without replacement in an ordered sequence. Here $n = 15$ and $r = 3$.

$$ {}_{15}P_3 = \frac{15!}{12!} = 15 \times 14 \times 13 = 2730. $$

**Example 4:** What is the sum of all the different 2-digit integers formed when selecting from the digits 1, 6 and 9 where no repeats are allowed?

**Solution:** There are ${}_3P_2 = 6$ possibilities:

List: 16, 19, 61, 69, 91, 96

Sum: $16 + 19 + 61 + 69 + 91 + 96 = \mathbf{352}$

**Theorem 1: Permutations of distinguishable objects**

In general, from $n$ **distinguishable** objects, $r$ are chosen **without replacement** in an **ordered** sequence then the number of permutations is

$$ {}_nP_r = n(n-1)\cdots(n-r+1) = \frac{n!}{(n-r)!}. $$

# Combinations - Unordered Selection without Replacement

A subset of $r$ objects selected **without** regard to **order** from a set of $n$ **distinguishable** objects is called a combination of $n$ objects taken $r$ at a time. The total number of such combinations is symbolized by

$$ {}_nC_r \qquad \text{or} \qquad \binom{n}{r}. $$

**Equivalent Problem:** allocate $r$ **indistinguishable** objects into $n$ **distinguishable** cells, with no cells containing $> 1$ objects.

**Example 5:**

What is the number of combinations of three letters chosen from four letters A, B, C and D?

| Combinations | Permutations |
|---|---|
| {A, B, C} | (A,B,C), (A,C,B), (B,A,C), (B,C,A), (C,A,B), (C,B,A) |
| {A, B, D} | (A,B,D), (A,D,B), (D,A,B), (D,B,A), (B,A,D), (B,D,A) |
| {A, C, D} | (A,C,D), (A,D,C), (C,A,D), (C,D,A), (D,A,C), (D,C,A) |
| {B, C, D} | (B,C,D), (B,D,C), (C,B,D), (C,D,B), (D,B,C), (D,C,B) |

Number of permutations $= {}_4P_3 = \dfrac{4!}{(4-3)!} = \dfrac{24}{1} = 24.$

Number of combinations $= {}_4C_3 = \dbinom{4}{3} = \dfrac{4!}{3!1!} = \dfrac{24}{6} = 4.$

List out the number of permutations and divide out the repeats.

In general, we have

$$ {}_nC_r = \begin{cases} \dfrac{n!}{r!(n-r)!}, & \text{for } r \le n; \\ 0, & \text{for } r > n. \end{cases} $$

Note: ${}_nC_r = {}_nC_{n-r}$ or $\binom{a+b}{a} = \binom{a+b}{b}$.

**Example 6:**

A beauty contest has 15 candidates. How many ways can 5 contestants be chosen to enter the final round?

**Solution:**

**Problem:** $n$ **distinguishable** objects, choose $r$ from them **without replacement** in an **unordered** sequence. Here $n = 15$ and $r = 5$; ${}_{15}C_5 = \dfrac{15!}{5!10!} = 3003$.

**Example 7:** In how many ways can a committee of 4 be selected from a group of 16 people?

**Solution:** ${}_{16}C_4 = \dfrac{16!}{4!12!} = 1820$.

**Example 8:**

How many combinations can be formed from six numbers chosen from $\{1, 2, \ldots, 49\}$ as in Mark Six?

**Solution:** ${}_{49}C_6 = \dfrac{49!}{6!43!} = 13983816$.

**Example 9:**

In order to play a game of basketball, 10 children at a playground divide themselves into two teams of 5 each. How many different divisions are possible?

**Solution:** $\dfrac{{}_{10}C_5}{2} = 126$.

**Example 10:**

In how many different ways can 20 true/false questions of a test be marked such that 7 are right and 13 are wrong?

**Solution:** ${}_{20}C_7 = {}_{20}C_{13} = 77520$.

# Selection of Distinguishable Objects

Both the formulae ${}_nP_r$ and ${}_nC_r$ described in previous sections can be used to count the number of subsets of size $r$ from a set of $n$ **distinguishable** objects, with ${}_nP_r$ as the number of ***ordered*** subsets and ${}_nC_r$ as the number of ***unordered*** subsets. One point to note is that in both cases, the subsets are selected **without replacement**, i.e., repeated objects are not allowed in the selection.

In some problems we may need to evaluate the number of subsets selected **with replacement**.

# Ordered Selection with Replacement

**Problem:** $n$ distinguishable objects. Choose $r$ from them **with replacement** in an **ordered** sequence.

**Equivalent Problem:** allocate $r$ **distinguishable** objects into $n$ **distinguishable** cells, with no restriction on the number of objects allocated in each cell.

**Example 11:**

The pass code of a number lock consists of 4 digits, each digit may be set to 0,1,2,. . . ,9.
How many possible pass codes are there?

**Solution:** The number of possible pass codes can be easily computed as

$$10 \times 10 \times 10 \times 10 = 10^4 = 10000.$$

It is a typical example of counting the number of *ordered* subsets (4 digits) from a set of distinguishable objects (0 – 9) **with replacement** as there can be repeated digits in the pass codes (e.g., 9901, 1535, etc.) and different orderings of the digits lead to different pass code (e.g., 1234 is different from 2314, 4478 is different from 7484, etc.).

**Theorem 2: Ordered selection with replacement**

In general, the number of **ordered** selections of $r$ out of $n$ **distinguishable** objects can be computed by $n^r$.

**Example 12:** How many arrangements are possible when rolling 3 fair dice of different colours?

**Solution:** $6 \times 6 \times 6 = 6^3 = 216$.

**Example 13:** A fair coin is tossed 4 times. How many possible sequences of outcomes are there?

**Solution:** $2 \times 2 \times 2 \times 2 = 2^4 = 16$.

**Example 14:** In an election there are 15 voters and 4 candidates. Suppose all votes are

valid. If the voters identity is listed on their ballot paper, how many different voting records are possible?

**Solution:** $4 \times 4 \times \cdots \times 4 = 4^{15} = 1073741824$.

# More Partitioning of Objects - Multinomial

**Theorem 4: Partitioning of distinguishable objects**

If a set of $n$ **distinct** objects is to be partitioned into $r$ **distinct** groups of respective size $n_1, n_2, \ldots, n_r$ where $n = n_1 + n_2 + \cdots + n_r$. The number of different partitions is

$$\binom{n}{n_1, n_2, \ldots, n_r} = \frac{n!}{n_1! n_2! \cdots n_r!}.$$

It is known as a **multinomial** coefficient.

**Example 15:** Consider the word "PEPPER". How many ways can the letters P, E, P, P, E, R be rearranged to form a 6-letter word?

**Solution:** $\dfrac{6!}{3!2!1!} = 60$ (3 L's, 2 E's and 1 R).

**Example 16:** How many different letter arrangements can be formed using the letters LETTERS?

**Solution:** $\dfrac{7!}{1!1!1!2!2!} = 1260$ (1 L, 1 R, 1 S, 2 E's, 2 T's).

**Example 17:**

A box contains four books on history and twelve books on mathematics. If the books are distributed equally at random among four students, find the probability that each student will get a book on history.

**Solution:**

Number of ways that each student will get a book on history is

$$\frac{4!}{1!1!1!1!} \times \frac{12!}{3!3!3!3!} = 8870400.$$

Number of ways of distributing the 16 books to the four students is

$$\frac{16!}{4!4!4!4!} = 63063000.$$

Hence, the required probability is $\dfrac{8870400}{63063000} = 0.1407$.

# Part B - Basic Probability Laws

## Probability

To most people, ***probability*** is a loosely defined term employed in everyday conversation to indicate the measure of one's belief in the occurrence of a future event. For example, what is the chance of arriving on time if one takes the bus? takes the taxi?

However for scientific purposes, it is necessary to give the word **probability** a definitive, clear interpretation. One can think of probability as the language when discussing uncertainty. For instance, in tossing a coin (random experiment), the outcomes can either be $\{H\}$ or $\{T\}$. If we know exactly what the outcome of the next trial will be, then we are certain that, say, a $\{H\}$ will show up next. However it is impossible to know exactly what is going to happen in reality - **uncertainty**.

**<u>Definition 1:</u>**

A **random experiment** is a process leading to at least two possible outcomes with uncertainty as to which will occur. Examples are (i) tossing a coin; (ii) rolling a die; (iii) daily changes in Hang Seng Index.

**<u>Definition 2:</u>**

The possible outcomes of a random experiment are called the **basic outcomes**, and the set of all basic outcomes is called the **sample space**.

Examples are (i) $\Omega = \{H, T\}$; (ii) $\Omega = \{1, 2, 3, 4, 5, 6\}$; and (iii) $\Omega = \{\text{up}, \text{down}\}$.

**<u>Definition 3:</u>**

An **event** is a set of basic outcomes, or a collection of some basic outcomes from the sample space, and it is said to occur if the random experiment gives rise to one of its constituent basic outcomes.

An example of (ii): An event can be $A = \{1, 3, 5\} = \{\text{the number rolled is odd}\}$

**Theorem 1:**

If an experiment can result in any one of $n$ different **equally likely** outcomes, and if $r$ of these outcomes together constitute event $A$, then

$$P(A) = \frac{r}{n}.$$

Comment: Frequentists view of probability - frequency interpretation of probability - **long run average**. More precisely, if the random experiment can be repeated over and over again with $n =$ total number of trials and $r =$ number of trials in which event $A$ actually happened out of the $n$ trials. Then $\Pr(A)$ is simply the proportion or rate of occurrences of event $A$ in the long run; or

$$P(A) \;=\; \frac{r}{n} \;=\; \text{relative frequency of event } A.$$

For example, $P(\text{new born baby is a male}) = 0.47$. Hence, probability is an abstraction of relative frequency; or the relative frequency is a realization of probability.

# Axioms (Postulates) of Probability

1. If $A$ is an event in the sample space $\Omega$, then

$$0 \leq P(A) \leq 1.$$

2. $P(\Omega) = 1$.

   The following properties are general results of the above axioms of probability:

3. $P(\phi) = 0$.
4. $P(E') = 1 - P(E)$ where $E'$ is the complement of $E$.
5. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
6. $$\begin{aligned} P(E_1 \cup E_2 \cup \cdots \cup E_k) = \; & \textstyle\sum_{i=1}^{k} P(E_i) - \sum\sum_{1 \leq i < j \leq k} P(E_i \cap E_j) \\ & + \sum\sum\sum_{1 \leq i < j < l \leq k} P(E_i \cap E_j \cap E_l) + \cdots \\ & + (-1)^{k+1} \; P(E_1 \cap E_2 \cap \cdots \cap E_k) \end{aligned}$$
7. $P(A' \cap B) = P(B) - P(A \cap B)$.

# Additive Law of Probability, Mutually Exclusive Events & Independence

## Additive Law of Probability

For events $A$ and $B$

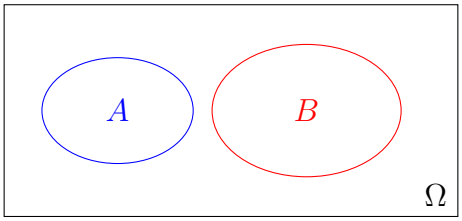
$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

This law is **always true**.

## DISJOINT or Mutually Exclusive

If events $A$ and $B$ are disjoint (mutually exclusive) then $A \cap B = \phi$.

Thus, $P(A \cap B) = P(\phi) = 0$.



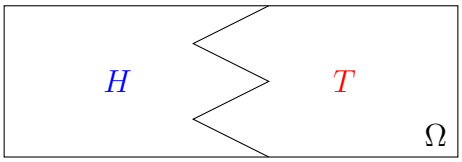

## Independence

If events $A$ and $B$ are independent then

$$P(A \cap B) = P(A) \cdot P(B).$$

## Example 18: Coin Tossing

Toss a single fair coin once. The possibilities are $H = \{\text{Heads}\}$ or $T = \{\text{Tails}\}$. The Venn diagram looks like this:




$P(H) = 0.5; \quad P(T) = 0.5; \quad P(H \cap T) = P(\phi) = 0;$

$P(H) \times P(T) = 0.5 \times 0.5 = 0.25 \neq 0.$

Therefore the events $H$ and $T$ are **disjoint/mutually exclusive** but NOT independent.

**<u>Example 19:</u> Cards**

One card is selected at random from a regular deck of 52 cards. Find the probability that the card selected in each of the following is:

**(a)** a queen (Q).

**Solution:**

| | A | K | Q | J | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ♠ | | | ★ | | | | | | | | | | |
| ♡ | | | ★ | | | | | | | | | | |
| ♣ | | | ★ | | | | | | | | | | |
| ♢ | | | ★ | | | | | | | | | | |

$$P(Q) = \frac{4}{52} = \frac{1}{13}$$

**(b)** a heart (♡)

**Solution:**

| | A | K | Q | J | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ♠ | | | | | | | | | | | | | |
| ♡ | ★ | ★ | ★ | ★ | ★ | ★ | ★ | ★ | ★ | ★ | ★ | ★ | ★ |
| ♣ | | | | | | | | | | | | | |
| ♢ | | | | | | | | | | | | | |

$$P(\heartsuit) = \frac{13}{52} = \frac{1}{4}$$

**(c)** the queen of hearts

**Solution:**

| | A | K | Q | J | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ♠ | | | | | | | | | | | | | |
| ♡ | | | ★ | | | | | | | | | | |
| ♣ | | | | | | | | | | | | | |
| ♢ | | | | | | | | | | | | | |

Note: the queen of hearts is a QUEEN AND a HEART

$$P(Q \text{ of } \heartsuit) = P(Q \cap \heartsuit) = \frac{1}{52}$$

**(d)** Either a Q or a ♡ or both

**Solution:**

| | A | K | Q | J | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ♢ | | | ★ | | | | | | | | | | |
| ♡ | ★ | ★ | ★ | ★ | ★ | ★ | ★ | ★ | ★ | ★ | ★ | ★ | ★ |
| ♣ | | | ★ | | | | | | | | | | |
| ♠ | | | ★ | | | | | | | | | | |

$$P(Q \cup \heartsuit) = P(Q) + P(\heartsuit) - P(Q \cap \heartsuit) = \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$$

**(e)** Either a Q or a ♡ but not both

**Solution:**

| | A | K | Q | J | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ♢ | | | ★ | | | | | | | | | | |
| ♡ | ★ | ★ | | ★ | ★ | ★ | ★ | ★ | ★ | ★ | ★ | ★ | ★ |
| ♣ | | | ★ | | | | | | | | | | |
| ♠ | | | ★ | | | | | | | | | | |

$$P(Q \text{ or } \heartsuit \text{ but NOT both}) = \frac{15}{52}$$

**(f)** Are the events Q and ♡ disjoint/mutually exclusive?

**Solution:**

$Q \cap \heartsuit \neq \phi$

Therefore, Q and ♡ are NOT disjoint/mutually exclusive.

**(g)** Are the events Q and ♡ independent?

**Solution:**

If Q and ♡ are INDEPENDENT events then $P(Q \cap \heartsuit) = P(Q) \times P(\heartsuit)$

Here, $P(Q \cap \heartsuit) = \frac{1}{52}$ and $P(Q) \times P(\heartsuit) = \frac{1}{13} \times \frac{1}{4} = \frac{1}{52}$

Therefore, Q and ♡ are independent events.

<u>**Conclusion**</u>: Q and ♡ are **independent** but NOT disjoint/mutually exclusive.

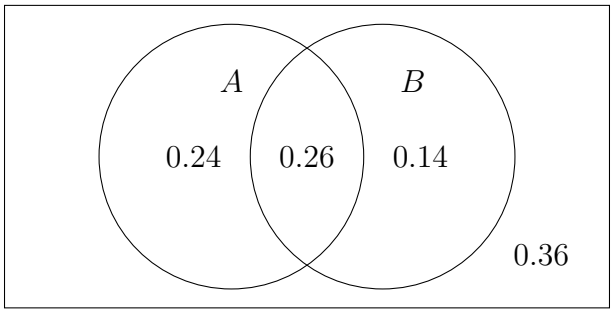
**<u>Example 20:</u> Library books**

Let $A$ be the event that a library book on statistics is presently checked out of the university library and let $B$ be the event that a second statistics book is checked out.

**(a)** If $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cup B) = 0.64$ then compute $P(A \cap B)$ and draw the Venn diagram.

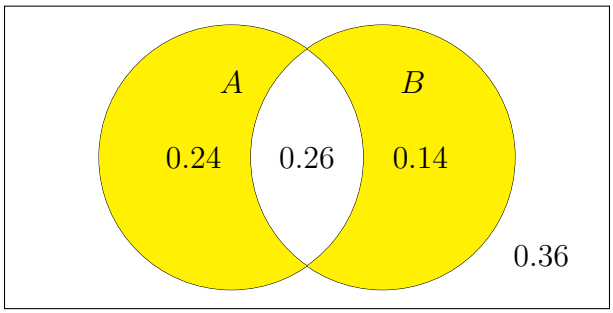
**Solution:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Therefore $P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.5 + 0.4 - 0.64 = 0.26$




**(b)** Compute the probability that exactly one of the two books is checked out.

**Solution:** We want $P(A \cap B') + P(A' \cap B) =?$




Looking at the above diagram we can see

$$P(A \cap B') + P(A' \cap B) = 0.24 + 0.14 = 0.38$$

Alternatively,

$$P(A \cap B') + P(A' \cap B) = P(A \cup B) - P(A \cap B) = 0.64 - 0.26 = 0.38$$

# Summary of Mutually Exclusive and Independent events

| | Disjoint/mutually exclusive | Independent |
|---|---|---|
| Coin tossing | ✓ | × |
| Cards | × | ✓ |
| Library books | × | × |
| $\emptyset$ empty set | ✓ | ✓ |

**Show that $A$ and $B$ cannot be both disjoint(exclusive) and independent.**

Suppose $0 < P(A) < 1$ and $0 < P(B) < 1$.

If $A$ and $B$ are independent then $P(A) \times P(B) = P(A \cap B)$

If $A$ and $B$ are mutually exclusive then $P(A \cap B) = P(\phi) = 0$

If both conditions hold then $P(A) \times P(B) = 0$

Either $P(A) = 0$ or $P(B) = 0$

Either $A = \phi$ or $B = \phi$

Therefore, $A$ and $B$ **CANNOT be both disjoint(exclusive) and independent**.

(Unless either of them is the empty set.)

# Conditional Probability

**<u>Example 21:</u> Police & burglar**

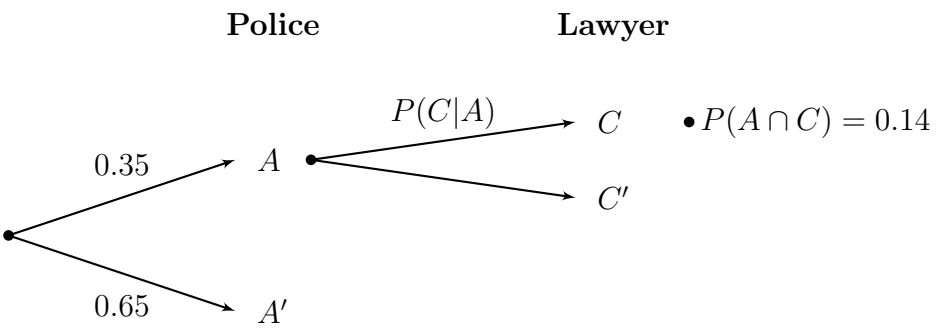
Police records show that for burglaries in a certain city the probability is 0.35 that an arrest will be made. The probability that an arrest and conviction will occur is 0.14. What is the probability that a person arrested for burglary will be convicted?

**Solution:**

Let $A$ = Arrest and $C$ = Conviction

From question : $P(A) = 0.35$; $P(A \cap C) = 0.14$

$$P(C \mid A) = \frac{P(A \cap C)}{P(A)} = \frac{0.14}{0.35} = 0.4$$




**<u>Example 22:</u> M.P.F. scheme**

The following **contingency table** was obtained for 15,000 employees of a large company having to join a mandatory provident fund (M.P.F.) scheme. The scheme has three options, namely low risk (A), moderate risk (B) and high risk (C) and the company has 9000 male (M) employees and 6000 female (F) employees.

| | Low (A) | Moderate (B) | High (C) | Total |
|---|---|---|---|---|
| Male (M) | 1500 | 3500 | 4000 | 9000 |
| Female (F) | 3000 | 2000 | 1000 | 6000 |
| Total | 4500 | 5500 | 5000 | 15000 |

An employee is selected at random.

**(a)** What is the probability that the employee chooses a high risk option?

$$P(\text{High}) = \frac{5000}{15000} = \frac{1}{3} = 0.333333$$

**(b)** What is the probability of being a male employee and choosing a high risk option?

$$P(\text{Male} \cap \text{High}) = \frac{4000}{15000} = \frac{4}{15} = 0.266667$$

**(c)** Given that the employee selected is female, what is the probability that she chooses a high risk M.P.F. option?

$$P(\text{High} \mid \text{Female}) = \frac{1000}{6000} = \frac{P(H \cap F)}{P(F)} = \frac{\frac{1000}{15000}}{\frac{6000}{15000}} = \frac{1}{6} = 0.16666667$$

**(d)** If it is known that an employee chooses the high risk option, what is the probability that this employee is female?

$$P(\text{Female} \mid \text{High}) = \frac{1000}{5000} = \frac{P(F \cap H)}{P(H)} = \frac{1}{5} = 0.2$$

# Bayes' theorem

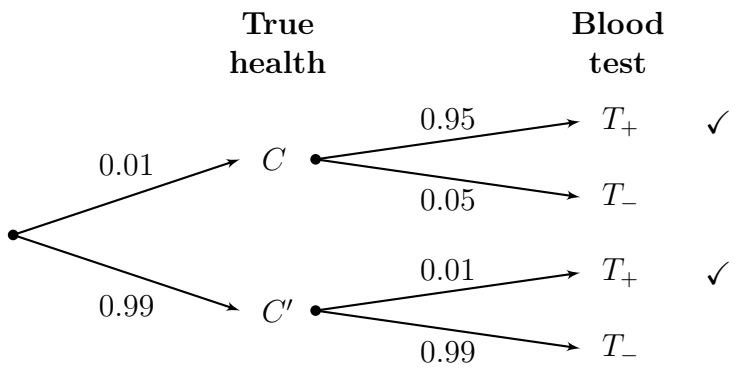
**<u>Example 23:</u> Cancer**

A test for detecting a certain type of cancer gives a positive result for 95% of a large number of patients subsequently found to have the cancer, and gives a negative result for 99% of those not having the cancer. If the test is applied randomly to a population in which the proportion of persons having the cancer is 1%, then find the probability that a person has the cancer if the test gave a positive result.

**Solution:**

Let $C$ = person has the cancer and $T_+$ = test result is positive.

From question: $P(T_+ \mid C) = 0.95$; $P(T_- \mid C') = 0.99$; $P(C) = 0.01$




By Bayes'theorem :

$$P(C|T_+) = \frac{P(C \cap T_+)}{P(T_+)} = \frac{0.01 \times 0.95}{0.01 \times 0.95 + 0.99 \times 0.01} = \frac{0.0095}{0.0194} = 0.48969$$

**<u>Example 24:</u> Car theft**

Records show that among all car theft cases in Hong Kong, 25% occurred on Hong Kong Island, 30% in Kowloon and 45% in the New Territories. The probabilities that a car stolen in these three districts can be recovered are, respectively, 0.45, 0.25 and 0.35.

**Solution:**

Let HK = car theft on Hong Kong Island

K = car theft in Kowloon

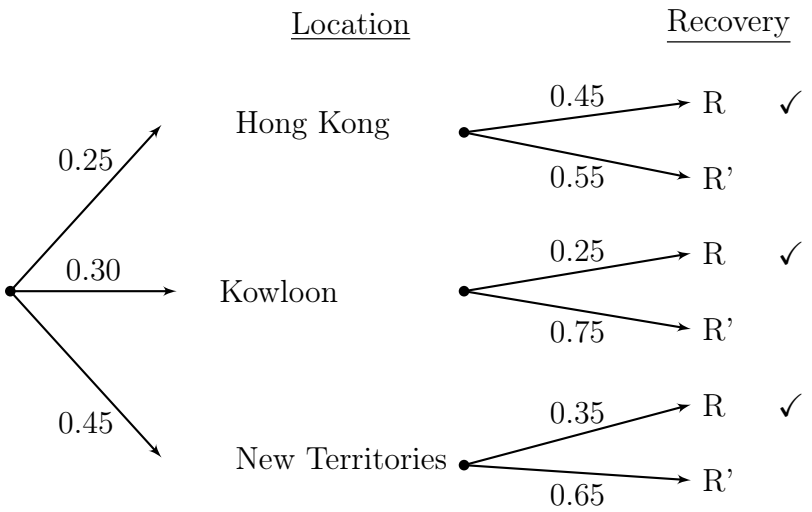
NT = car theft in the New Territories

R = stolen car is recovered

From the question: $P(HK) = 0.25, \quad P(K) = 0.30, \quad P(NT) = 0.45,$

$P(R \mid HK) = 0.45, \quad P(R \mid K) = 0.25, \quad P(R \mid NT) = 0.35$

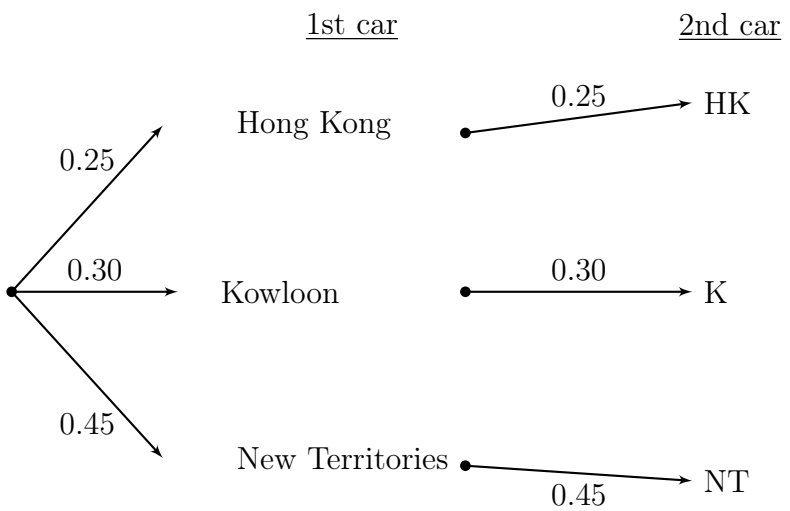
**(a)** Find the probability that a recovered car was stolen in Kowloon.




By Bayes' theorem:

$$P(K \mid R) = \frac{P(K \cap R)}{P(R)} = \frac{0.3 \times 0.25}{0.25 \times 0.45 + 0.3 \times 0.25 + 0.45 \times 0.35} = \frac{0.075}{0.345} = 0.21739$$

**(b)** Suppose that two cars were stolen and that the two cases were independent of each other. Find the probability that the two cases occurred in the same district.




$P(HK^2 + K^2 + NT^2) = 0.25^2 + 0.3^2 + 0.45^2 = 0.355$

**<u>Example 25:</u> Equally Likely**

**(a)** Albert draw cards from an ordinary deck of 52 cards without replacement. What is the probability that Albert gets a Queen on his 10th draw?

**Solution:** $P(\text{Q on 10th draw}) = P(\text{Q on 1st draw}) = \dfrac{4}{52} = \dfrac{1}{13}$

**(b)** Albert and Bing take turns to draw a card from an ordinary deck of 52 cards with Albert first. What is the probability that Bing gets a King on his 12th draw?

**Solution:** $P(\text{K on 12th draw}) = P(\text{K on 1st draw}) = \dfrac{4}{52} = \dfrac{1}{13}$

**(c)** Albert and Bing take turns to draw a card from an ordinary deck of 52 cards with Albert first drawing the Ace of Clubs. What is the probability that Bing draws a King on his next draw?

**Solution:** $P(\text{King} \mid \text{Ace of Clubs}) = \dfrac{4}{51}$

**<u>Example 26:</u> Marbles in an urn**

An urn contains 3 red and 2 white marbles. A person randomly selects 4 marbles one after the other, without replacement. What is the probability that the first and the last marbles picked are both white?

**Solution:**

$$P(\text{1st W and last W}) = P(\text{1st W and 4th W}) = P(WRRW) = \frac{2}{5} \times \frac{1}{4} = \frac{2}{20} = \frac{1}{10} = 0.1$$

**Explanation:**

The middle terms all cancel.

$$P(WRRW) = \frac{2}{5} \times \frac{3}{4} \times \frac{2}{3} \times \frac{1}{2} = \frac{2}{5} \times \frac{1}{4} = 0.1$$

**<u>Example 27:</u> More marbles in an urn**

An urn contains 7 red and 3 white marbles. Five marbles are drawn one by one, without replacement. What is the probability that the first and the last marbles picked are both red?

**Solution:**

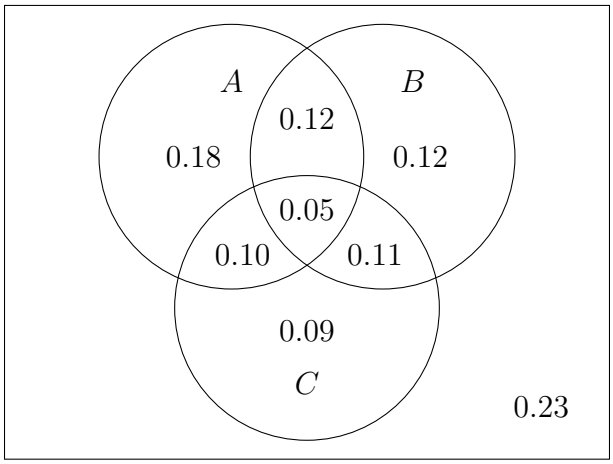
$$P(\text{1st R and last R}) = P(\text{1st R and 5th R}) = P(R???R) = \frac{7}{10} \times \frac{6}{9} = \frac{42}{90} = \frac{7}{15} = 0.466666667$$

# Extension of Additive Law of Probability to three events

**Example 28:**

Suppose that $P(A) = 0.45$, $P(B) = 0.4$, $P(C) = 0.35$, $P(A \cap B) = 0.17$, $P(B \cap C) = 0.16$, $P(A \cap C) = 0.15$ and $P(A \cap B \cap C) = 0.05$.

**(a)** Draw the Venn diagram.




**(b)** Find $P(A \cup B)$ and $P(A \cup B \cup C)$.

**Solution:**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.45 + 0.4 - 0.17 = 0.68$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

$$\therefore P(A \cup B \cup C) = 0.45 + 0.4 + 0.35 - 0.17 - 0.16 - 0.15 + 0.05 = 0.77$$